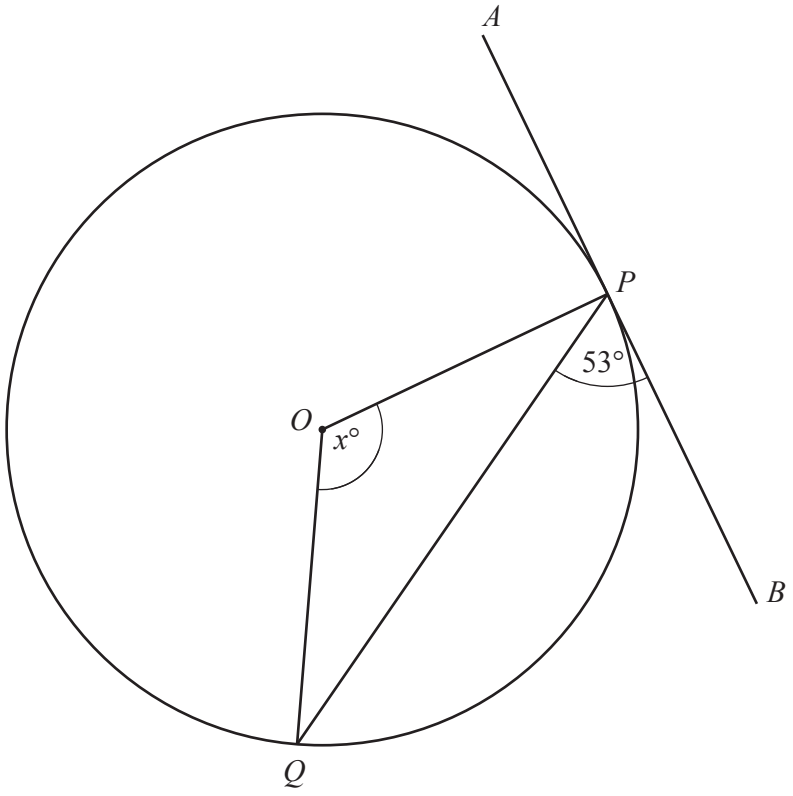
0580_w20_qp_33_Q8

8 (a)



P and *Q* are points on the circle, centre *O*.
APB is a tangent to the circle at *P*.

(i) Write down the mathematical name for the line *PQ*.

..... [1]

(ii) Explain why angle *OPB* is 90°.

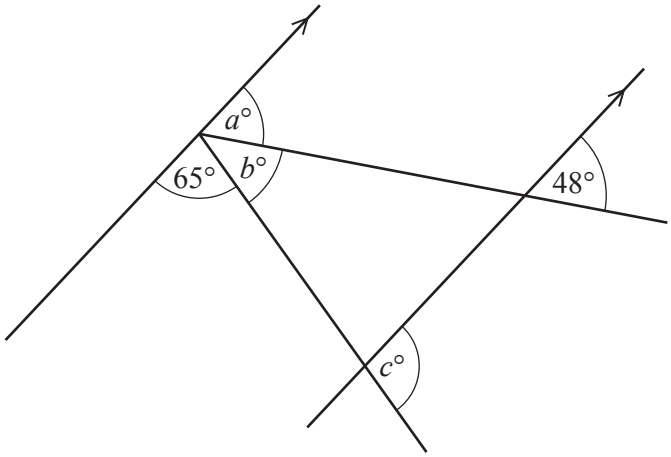
..... [1]

(iii) Find the value of *x*.

$x =$ ..... [3]

0580_w20_qp_33_Q8

(b)



NOT TO SCALE

The diagram shows two parallel lines and two straight lines.

(i) Find the value of a.
Give a reason for your answer.

a = ... because ..... [2]

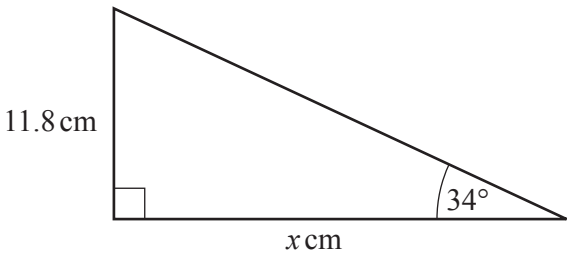
(ii) Find the value of b.
Give a reason for your answer.

b = ... because ..... [2]

(iii) Find the value of c.

c = ... [2]

(c)



NOT TO SCALE

Calculate the value of x.

x = ... [3]